

SDN CONTROLLER EVALUATION

Robust Network Virtualization & Multi-Tenancy Capabilities

EXECUTIVE SUMMARY

For a company requiring robust network virtualization and isolated multi-tenant virtual networks over shared physical infrastructure, **VMware Nicira (NVP / NSX)** is the most relevant and powerful solution among the listed options.

1. VMWARE NICIRA (NVP / NSX) ANALYSIS

Nicira (pioneer of NVP, Network Virtualization Platform, which evolved into VMware NSX) was purpose-built to solve the multi-tenant network virtualization challenge in cloud datacenters.

- **Overlay-Based Virtualization:** Nicira creates completely decoupled logical network topologies (L2 switches, L3 routers, firewalls, load balancers) in software over any IP transport network using overlay encapsulation protocols (such as STT, VXLAN, and NVGRE).
- **Complete Tenant Isolation:** It supports multi-tenancy at scale, allowing overlapping IP addresses, tenant-specific security policies, and isolated virtual networks without requiring reconfiguration of physical underlying switches.
- **Hardware Independence:** Physical switches only need to forward standard IP packets, shielding the physical network from MAC table limits and VLAN ID scaling restrictions (4096 max).

2. ALTERNATIVE OPTIONS EVALUATION

- **Generic OpenFlow-Related Controllers:** While OpenFlow can enforce path isolation via flow tables, it operates primarily at the physical flow level and lacks native overlay orchestration, automated L2/L3 logical abstraction, and cloud platform management out of the box.
- **Plexxi:** Plexxi focuses on dynamic optical datacenter fabrics, photonic switching, and workload affinity optimization rather than multi-tenant software overlay virtualization.
- **Cisco onePK:** Cisco onePK is a programmatic SDK for inspecting and configuring Cisco router/switch hardware features natively, rather than a full network virtualization platform designed for multi-tenant software logical overlays.

CONCLUSION

VMware Nicira is explicitly designed for creating and managing programmatically isolated, software-defined virtual networks over shared physical infrastructure, making it the most relevant choice.